MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER

Discipline Specific Core Course (DSC)-MG3DSCCSC200- Database Management Systems

(2024 ADMISSION ONWARDS)

Duration : 1.5 Hours **Maximum Marks : 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create ©, Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer All Questions

Each question carries **2** marks

1. Define database and DBMS. [U] [1]
2. List the advantages of using DBMS approach [U] [1]
3. Why is entity integrity important in a relational database? [U] [2]
4. Define normalization. [U] [2]
5. State the purpose of adding default constraint to a table in a relational database. [U][3].

(2x5=10 marks)

Part B

Short Answer Questions

Answer any **five** questions

Each question carries **5** marks

6. Explain the functions of DBMS. [U] [1]
7. Describe the three-schema architecture of DBMS. [U] [1]

8. Explain briefly about the different types of users in DBMS. [U] [1]
9. State the characteristics of relations in relational model. [U] [2]
10. What is functional dependency? Explain with an example. [U] [2]
11. List and explain the various aggregate functions in SQL with examples [A] [3]
12. Write SQL commands to create a table STUDENT with attributes ID, Name, Age, Course. [A] [3]

Part B

Essay Questions

Answer any **1** question

Each question carries **15** marks

13. Draw an ER diagram for a Library Management System with entities: Book, Author, Member, and Issue. Include appropriate attributes and relationships. [A] [1]
14. Discuss the Relational Data Model. Explain:
    a. Basic concepts of relational model (5 marks)
    b. Relational model constraints (5 marks)
    c. Normalization up to Third Normal Form with examples (5 marks)

    [An] [2]